Customer Churn Prediction Using Machine Learning

Introduction

Customer churn refers to the loss of customers who stop using a company's products or services. Predicting churn helps organizations identify at-risk customers and implement retention strategies. This project develops a machine learning model to predict customer churn using customer demographic, service, and billing information.

Objective

The objective of this project is to build an end-to-end machine learning solution capable of predicting whether a customer is likely to churn. The project includes data cleaning, visualization, preprocessing, model training, evaluation, error analysis, and business insights.

Dataset Description

The Telco Customer Churn dataset contains customer information such as:

- Gender
- Senior Citizen Status
- Partner Status
- Dependents
- Tenure
- Internet Service
- Contract Type
- Payment Method
- Monthly Charges
- Total Charges
- Churn Status

Target Variable:

- Churn (Yes/No)

Data Cleaning

The following preprocessing steps were performed:

- Removed the customerID column.
- Converted TotalCharges to numeric values.
- Replaced missing values using the median.
- Removed duplicate records.
- Encoded categorical variables into numerical format.

Exploratory Data Analysis

Several visualizations were generated to understand the dataset:

Churn Distribution

Shows the proportion of customers who churned versus those who remained.

Contract Type Distribution

Illustrates the frequency of different contract types among customers.

Tenure Distribution

Shows how long customers stayed with the company.

Feature Engineering and Preprocessing

The dataset was transformed into a machine-learning-ready format through:

- Label Encoding
- Feature Selection
- Train-Test Split (80:20)

Model Training

Two machine learning models were trained and compared:

Logistic Regression

A simple and interpretable baseline classification model.

Random Forest

An ensemble model combining multiple decision trees for improved predictive performance.

Evaluation Metrics

The following metrics were used:

Accuracy

Measures the percentage of correctly classified customers.

Precision

Measures how many predicted churn cases were actually churn cases.

Recall

Measures how many actual churn cases were identified.

F1 Score

Provides a balance between precision and recall.

Confusion Matrix

Visualizes correct and incorrect classifications.

Model Comparison

The performance of Logistic Regression and Random Forest models was compared. The model with the highest predictive performance was selected as the final model.

Error Analysis

Misclassified customer records were analyzed to understand prediction failures. Some customers exhibited characteristics similar to both churned and retained customers, making classification more challenging.

Business Insights

The analysis revealed several important business insights:

1. Customers with shorter tenure are more likely to churn.
2. Month-to-month contracts have higher churn rates.
3. Customers with higher monthly charges show increased churn tendencies.
4. Long-term contracts improve customer retention.
5. New customers should be prioritized in retention campaigns.

Conclusion

A complete machine learning pipeline was successfully developed to predict customer churn. The selected model demonstrated strong predictive performance and generated actionable business insights. Such a solution can help organizations proactively identify at-risk customers and improve retention strategies.